RESEARCH OVERVIEW / VERSION 1.0.0

Structural Intelligence Framework

An Open Architecture for Intelligence Localization, Composition, Governance, and Structural Growth

Authors: Sizhe Tan & GPT-Obot Release: v1.0.0 License: Apache-2.0

Core Idea

SIF proposes an open, evidence-governed architecture in which intelligence capabilities are located, composed, challenged, reused, and revised rather than treated as a fixed collection of models. Structural precedents connect problem requirements to available capabilities; explicit contracts and relay interfaces support heterogeneous components; adversarial validation and residual analysis inform possible structural growth. The framework is a research proposal, not a validated general intelligence system.

Four Functional Domains

01 Localization	Address problems, contexts, capabilities, and execution paths (LLM, CCC, MDT, two-phase search, relay, CallingGraph).
02 Per-Node Intelligence	Compose heterogeneous methods, including LLMs, rules, optimization, simulation, ANN, and domain-specific tools.
03 Control & Scoring	Separate policy/authority, scoring/decision, and evidence/audit; support scoring trees, forests, and folding.
04 Structural Growth	Use residuals, counter-evidence, and governed experiments to propose, validate, adapt, or retire structures.

Connecting Mechanisms

Dual maps: Structural Problem-Solving Map ↔ Structural Capability Map, linked through bidirectional mapping (BSIM). Precedents: registry of capability definitions, cases, evidence, counter-evidence, and applicability boundaries. Interoperability: Structural Intelligence Contracts (SIC), Universal Naming, Typing and Relay (UNT-R), portable PIRP/PIRU units. Validation: Multi-Intelligence Adversarial Sandbox (MIAS) with independent evaluator challenge and separate action authorization.

Research Package & Navigation

18 research manuscripts across Foundations (3), Functional Domains (4), Interoperability (4), Validation and Evolution (3), and Frontiers (4). 4 candidate specifications covering SIC, UNT-R API, precedent registry schema, and sandbox evidence/growth lifecycle. 3 research resources: Open Questions, Testable Hypotheses, and Validation Roadmap. 8 architecture figures are planned.

PROJECT REPOSITORY / DOCUMENTS AND LINKS

https://github.com/sizhet/Structural-Intelligence-Framework

Start with README.md, START-HERE.md, and CONTENTS.md. Citation metadata: CITATION.cff and .zenodo.json. DOI: to be added after archival assignment.

Research status: Conceptual architecture and candidate engineering specifications. Claims about scaling, structural growth, perception horizons, and perspective-relative ASI remain hypotheses requiring controlled experiments and independent replication.